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Developing amharic text-based chatbot model for HIV/AIDS awareness and care using deep learning approaches

Abebe Sintayehu¹ and Eshete Derb Emiru^{2*}

Abstract

Background The healthcare sector relies heavily on human interaction, yet conversational AI applications like chatbots are becoming increasingly important. While global standard HIV/AIDS patient assistance conversational AI bots exist, they often lack compatibility with diverse languages and the specific contexts of users' living standards in various countries. This study proposes an Amharic text-based chatbot model designed for HIV/AIDS awareness and care advising, utilizing deep learning algorithms.

Results We collected and prepared 10,291 text data entries from reliable sources covering common HIV/AIDS topics, converting them to a semi-structured JSON format. After preprocessing through cleaning, normalization, tokenization, and embedding, FastText was used for feature extraction. We evaluated different deep learning algorithms, including LSTM, BiLSTM, GRU, and BiGRU, using accuracy, precision, recall, F1-score, and human evaluation. Extensive experiments were conducted, and the Adaptive Moment Estimation (Adam) optimizer outperformed the Stochastic Gradient Descent (SGD) optimizer. To prevent overfitting, dropout and early stopping techniques were employed. The BiGRU model achieved the highest performance, with a testing accuracy of 95.01% and a loss of 0.372, based on an 80:20 train-test split, a batch size of 32, 50 epochs, and a learning rate of 0.0001.

Conclusions The developed Amharic text-based chatbot model demonstrates promising performance in providing HIV/AIDS awareness and care advice, with the BiGRU model proving to be the most effective algorithm for this task. This model has the potential to improve access to vital health information for Amharic speakers, addressing linguistic and contextual barriers present in existing global solutions.

Keywords Deep learning, Chatbot, Amharic text, LSTM, GRU, BiGRU

Introduction

Human Immunodeficiency Virus (HIV) and Acquired Immunodeficiency Syndrome (AIDS) remain significant global health challenges, impacting millions of people

around the world. HIV weakens the immune system by specifically targeting CD4 cells (T cells), which play a vital role in defending the body against infections [1]. Since HIV/AIDS was discovered in the early 1980 s, it has greatly affected public health, social structures, and economic progress. Although there have been improvements in treatment and prevention methods, the virus remains a major global challenge, especially in sub-Saharan Africa and other areas with limited resources [2]. HIV/AIDS remains a significant public health issue in Ethiopia, with far-reaching social, economic, and health consequences. Since the mid-1980 s, when the first cases were reported,

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the epidemic has faced persistent challenges, including inadequate healthcare infrastructure, cultural stigmas, and socioeconomic inequalities. According to estimates from the Ethiopian Public Health Institute (EPHI) for 2022 and 2023, there are 610,350 people living with HIV in the country, with 8,257 new infections and 11,322 AIDS-related deaths occurring annually [1].

In recent years, advancements in machine learning and artificial intelligence (AI) have revolutionized various sectors, particularly healthcare [3, 4]. Chatbots have emerged as vital tools for delivering accessible and timely healthcare information and support [5]. Powered by deep learning algorithms, these chatbots can engage users in natural language, offering personalized responses and recommendations based on extensive datasets and learned patterns [6]. As a result, they hold promise for enhancing healthcare communication and improving patient outcomes in diverse populations. Ethiopia has made significant strides in addressing HIV/AIDS through strategic national initiatives bolstered by international partnerships and funding [7]. These efforts have primarily focused on prevention, treatment, and care, aimed at reducing transmission rates and enhancing the quality of life for those affected [8]. Nevertheless, despite these advancements, challenges remain, largely due to the limited awareness and understanding of HIV and its prevention among the population [9]. Consequently, this study aims to assess the knowledge and attitudes of urban adults in Ethiopia towards HIV/AIDS, along with identifying the factors influencing these perceptions [10].

The complexities of the Amharic language present significant obstacles for developing effective natural language processing (NLP) models capable of understanding and generating Amharic text, particularly in medical contexts. The lack of comprehensive, annotated healthcare datasets in Amharic further hinders the creation of NLP solutions that can deliver accurate information regarding HIV/AIDS prevention and treatment [11]. Current chatbot solutions primarily cater to English-speaking users, often neglecting the linguistic and cultural nuances essential for Amharic speakers. This gap is concerning in Ethiopia, where insufficient access to culturally relevant HIV/AIDS information perpetuates stigma and misinformation about the disease. The proposed research seeks to fill this void by developing an Amharic chatbot, employing deep learning techniques to provide accurate and culturally-contextual information on HIV/AIDS, ultimately empowering individuals to make informed health decisions and improving public health outcomes.

Our study has the following contributions:

- We prepared a dataset comprising 10,291 text entries related to HIV/AIDS symptoms, causes, preven-

tion, and treatment was compiled in a JSON format, which underwent thorough preprocessing using natural language processing techniques for consistency and normalization.

- We developed an Amharic text-based chatbot model for HIV/AIDS awareness and care advice using LSTM, BiLSTM, GRU, and BiGRU deep learning techniques by a prepared dataset.
- Among the proposed deep learning algorithms, BiGRU model achieved the highest testing accuracy of 95.01% through optimal training parameters and the application of techniques to mitigate overfitting.
- The performance of the chatbot was assessed through human evaluation, demonstrating its capability to accurately understand user intents and provide appropriate responses, thus effectively assisting individuals affected by HIV/AIDS in the Amharic language.

Research methodology

The proposed architecture begins by loading a dataset prepared in JSON format. After loading, the next step is to preprocess the dataset, which includes text normalization, cleaning, and tokenization, all of which play a key role. These steps prepare the training data by converting it into a list of words (tokens). Once preprocessing of the dataset is completed, the dataset is split for training and testing; after that, feature extraction or word embedding is performed to transform the text into a matrix (vector) of features needed by machine learning algorithms.. After preprocessing, the user input is transformed into a numeric vector using the word embedding algorithm. The model is developed using LSTM, BiLSTM, GRU and BiGRU algorithms. The classification is done by taking the user input represented by word embedding model and comparing with the data we have in our model. It is controlled by threshold value i.e., we have the result if it is greater than the threshold value. Finally, the performance of the model evaluated in various evaluation metrics. The general proposed architecture of the model is given in the following Fig. 1 and the detail descriptions of each stage are included in the following subsections.

Data collection

To develop the proposed Amharic text-based chatbot model, the primary task was data collection. For this purpose, we gathered data from various reliable and accessible platforms. The corpus for our research was collected from several online websites, including the World Health Organization (www.who.int) and the World Health Line (www.healthline.com). We also referenced various books relevant to this field [12]. Furthermore, we compiled frequently asked questions from these online sources and

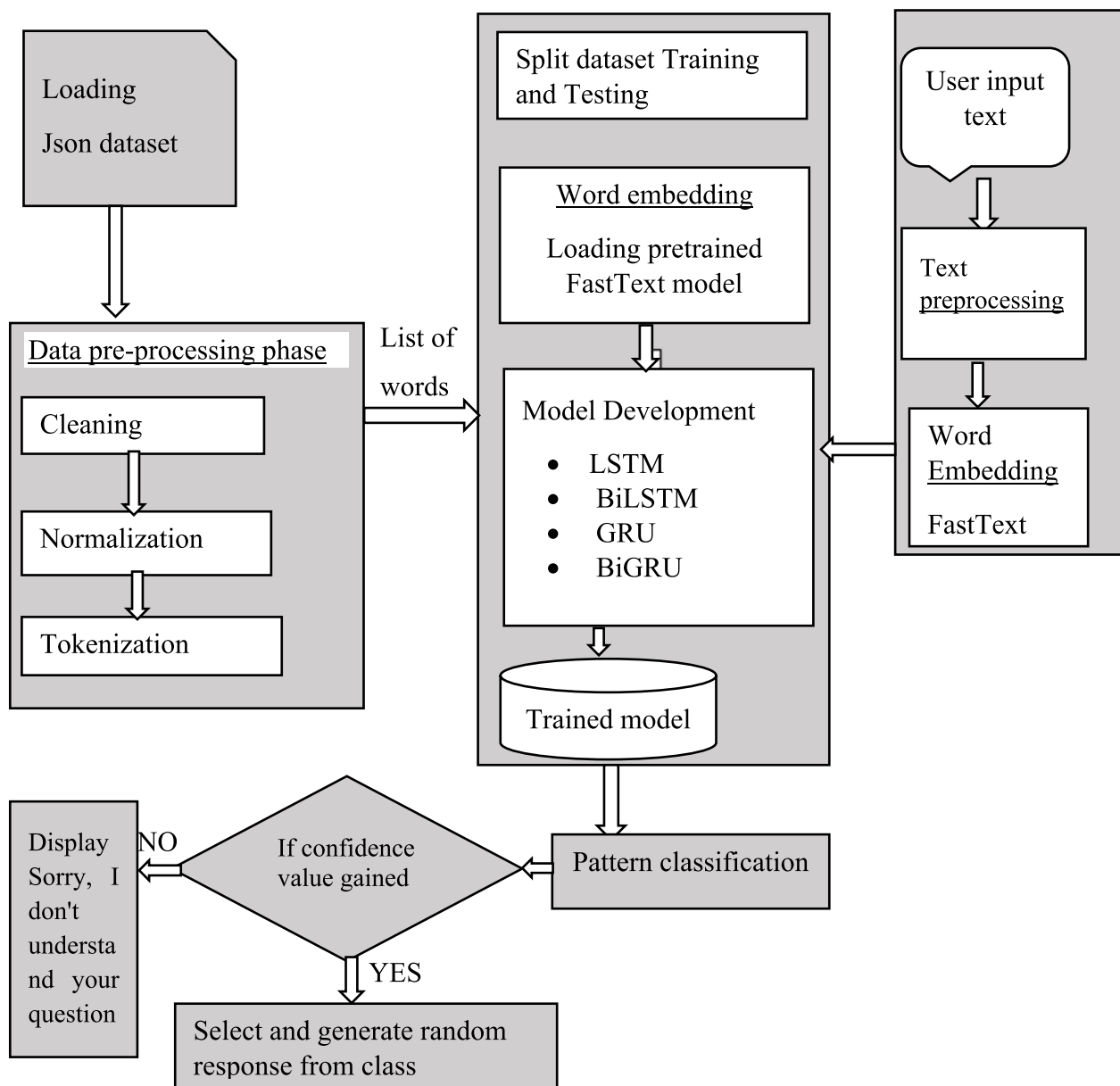


Fig. 1 Proposed architecture for amharic text-based chatbot model

conducted interviews with healthcare professionals to create a labeled dataset for training the deep learning model. The data is initially extracted into question-and-answer pairs to prepare it for the model. The model then learns what information to retain and how to extract relevant details. The utterances provided by different users vary, making the questions or intents crucial for guiding the model in remembering specific facts. This dynamic nature of the questions is captured using sequence-to-sequence modeling techniques, which account for the input structure and use Amharic word vectors for any

unseen words related to HIV/AIDS during training. Based on this dataset, the model is trained to predict the appropriate tag, randomly select a response, and provide the appropriate answer.

Dataset preparation

After collecting our data, the next step was to prepare the dataset for training the machine, which proved to be challenging. One primary issue was that the dataset obtained from the websites was in English, while we required an Amharic dataset. To address this, we first

translated the entire dataset from English to Amharic using Google Translate. Once the translation was complete, we organized the dataset into a semi-structured format using JSON [13]. This structure included three key components: tags, patterns, and responses.

- **Tag:** The intent or idea of the user's query, which is used as the target class while training the model.
- **Patterns:** The set of queries requested or frequently asked questions that are related to the intents.
- **Responses:** The set of possible responses related to the users' queries.

The following illustrates how we prepared the dataset. For instance, users may want to be greeted in Amharic, and they might ask, "ሁሉ ሰላም," "ሰላም ነው," "ፒስ ነው," "ሀይ," "ሰላም ነው ከይደደ," or "ከንዴት ነህ?" The responses to these queries could be "ሰላም ምን ልርዳዎት," "ጤና ይስጥልኝ," or "ደህና ነኝ." Thus, the intent or tag for this section of our dataset is "ሰላምታ." Therefore, the converted JSON dataset was structured accordingly.

Data preprocessing

Data preprocessing is a key task in many machine learning applications, particularly in natural language processing [14]. It plays a vital role in transforming text into a format that is more manageable for machine learning algorithms, enhancing their performance. This step ensures that the data is clean and structured, allowing algorithms to process it more effectively. Natural language processing (NLP) text preprocessing is the process of cleaning, removing noisy, normalizing and making complete and consistent data for the sake of develop the model [15]. Since, in NLP there are different text preprocessing tasks i.e., text cleaning, normalization, and tokenization, etc. The following preprocessing techniques are applied to develop the Amharic text based Chatbot for HIV/AIDS to assist HIV/AIDS positive persons and other individuals or health care professionals.

Text normalization and cleaning

Text normalization involves a series of related tasks aimed at standardizing text to ensure uniformity [16]. In the case of Amharic, normalization is influenced by the language’s writing system. The existence of multiple Fidel (symbols) that represent the same sound poses challenges in developing Amharic text-based models, such as chatbots. These different symbols must be considered as equivalent since they do not cause the meaning in the model. While these character variations may hold significance from a linguistic perspective, it is essential to normalize them when creating a text-based chatbot for the Amharic language. Otherwise,

the spelling variations of a word could unnecessarily increase the number of words representing an intent, which in turn could reduce the efficiency and accuracy of the model.

In this study, the solution to the problem involves selecting a single letter from groups of letters that share the same sound and replacing the others with it. For example, the letters ሀ, ሐ, ሑ, ሒ, and ሓ, as well as ሔ, ሕ, and ሖ and ሗ, as well as መ and ሙ, represent characters that have the same meaning and pronunciation but are represented by different symbols. This character variation also exists in many Amharic texts. To address this, these characters should be normalized to a single representative character; for instance, ሖ should be replaced with ሐ, ሐ with ሔ, and መ with ሙ, along with their respective variations (e.g., transforming ሙ, ሙ፡, ሙ፡፡, etc., to ሐ, ሐ፡, ሐ፡፡, etc.). For example, in the word "ጸረ ኤፍክደሺ," the letter "ጸ" is substituted with "ፀ" because both letters produce the same sound. This normalization process helps the computer to understand the word "ጸረ ኤፍክደሺ" has a similar representation with "ፀረ-ኤፍክደሺ". Another pre-processing step involved cleaning the text by removing different symbols and numbers from the text to make the computer understand the natural language easily. The Amharic language has its own numbering system, which includes characters like [፩, ፪, ፫, ፬, ፭, ፮, ፯, ፰, ፱, ፲, ፳, ፴, ፵, ፶, ፷, ፸, ፹, ፺, ፻, ፺፩, ፺፪, ፺፫, ፺፬, ፺፭, ፺፮, ፺፯, ፺፰, ፺፱, ፺፺, ፺፻, ፺፺፩, ፺፺፪, ፺፺፫, ፺፺፬, ፺፺፭, ፺፺፮, ፺፺፯, ፺፺፰, ፺፺፱, ፺፺፺, ፺፺፻, ፺፺፺፩, ፺፺፺፪, ፺፺፺፫, ፺፺፺፬, ፺፺፺፭, ፺፺፺፮, ፺፺፺፯, ፺፺፺፰, ፺፺፺፱, ፺፺፺፺, ፺፺፺፻, ፺፺፺፺፩, ፺፺፺፺፪, ፺፺፺፺፫, ፺፺፺፺፬, ፺፺፺፺፭, ፺፺፺፺፮, ፺፺፺፺፯, ፺፺፺፺፰, ፺፺፺፺፱, ፺፺፺፺፺, ፺፺፺፺፻, ፺፺፺፺፺፩, ፺፺፺፺፺፪, ፺፺፺፺፺፫, ፺፺፺፺፺፬, ፺፺፺፺፺፭, ፺፺፺፺፺፮, ፺፺፺፺፺፯, ፺፺፺፺፺፰, ፺፺፺፺፺፱, ፺፺፺፺፺፺, ፺፺፺፺፺፻, ፺፺፺፺፺፺፩, ፺፺፺፺፺፺፪, ፺፺፺፺፺፺፫, ፺፺፺፺፺፺፬, ፺፺፺፺፺፺፭, ፺፺፺፺፺፺፮, ፺፺፺፺፺፺፯, ፺፺፺፺፺፺፰, ፺፺፺፺፺፺፱, ፺፺፺፺፺፺፺, ፺፺፺፺፺፺፻, ፺፺፺፺፺፺፺፩, ፺፺፺፺፺፺፺፪, ፺፺፺፺፺፺፺፫, ፺፺፺፺፺፺፺፬, ፺፺፺፺፺፺፺፭, ፺፺፺፺፺፺፺፮, ፺፺፺፺፺፺፺፯, ፺፺፺፺፺፺፺፰, ፺፺፺፺፺፺፺፱, ፺፺፺፺፺፺፺፺, ፺፺፺፺፺፺፺፻, ፺፺፺፺፺፺፺፺፩, ፺፺፺፺፺፺፺፺፪, ፺፺፺፺፺፺፺፺፫, ፺፺፺፺፺፺፺፺፬, ፺፺፺፺፺፺፺፺፭, ፺፺፺፺፺፺፺፺፮, ፺፺፺፺፺፺፺፺፯, ፺፺፺፺፺፺፺፺፰, ፺፺፺፺፺፺፺፺፱, ፺፺፺፺፺፺፺፺፺, ፺፺፺፺፺፺፺፺፻, ፺፺፺፺፺፺፺፺፺፩, ፺፺፺፺፺፺፺፺፺፪, ፺፺፺፺፺፺፺፺፺፫, ፺፺፺፺፺፺፺፺፺፬, ፺፺፺፺፺፺፺፺፺፭, ፺፺፺፺፺፺፺፺፺፮, ፺፺፺፺፺፺፺፺፺፯, ፺፺፺፺፺፺፺፺፺፰, ፺፺፺፺፺፺፺፺፺፱, ፺፺፺፺፺፺፺፺፺፺, ፺፺፺፺፺፺፺፺፺፻, ፺፺፺፺፺፺፺፺፺፺፩, ፺፺፺፺፺፺፺፺፺፺፪, ፺፺፺፺፺፺፺፺፺፺፫, ፺፺፺፺፺፺፺፺፺፺፬, ፺፺፺፺፺፺፺፺፺፺፭, ፺፺፺፺፺፺፺፺፺፺፮, ፺፺፺፺፺፺፺፺፺፺፯, ፺፺፺፺፺፺፺፺፺፺፰, ፺፺፺፺፺፺፺፺፺፺፱, ፺፺፺፺፺፺፺፺፺፺፺, ፺፺፺፺፺፺፺፺፺፺፻, ፺፺፺፺፺፺፺፺፺፺፺፩, ፺፺፺፺፺፺፺፺፺፺፺፪, ፺፺፺፺፺፺፺፺፺፺፺፫, ፺፺፺፺፺፺፺፺፺፺፺፬, ፺፺፺፺፺፺፺፺፺፺፺፭, ፺፺፺፺፺፺፺፺፺፺፺፮, ፺፺፺፺፺፺፺፺፺፺፺፯, ፺፺፺፺፺፺፺፺፺፺፺፰, ፺፺፺፺፺፺፺፺፺፺፺፱, ፺፺፺፺፺፺፺፺፺፺፺፺, ፺፺፺፺፺፺፺፺፺፺፺፻, ፺፺፺፺፺፺፺፺፺፺፺፺፩, ፺፺፺፺፺፺፺፺፺፺፺፺፪, ፺፺፺፺፺፺፺፺፺፺፺፺፫, ፺፺፺፺፺፺፺፺፺፺፺፺፬, ፺፺፺፺፺፺፺፺፺፺፺፺፭, ፺፺፺፺፺፺፺፺፺፺፺፺፮, ፺፺፺፺፺፺፺፺፺፺፺፺፯, ፺፺፺፺፺፺፺፺፺፺፺፺፰, ፺፺፺፺፺፺፺፺፺፺፺፺፱, ፺፺፺፺፺፺፺፺፺፺፺፺፺, ፺፺፺፺፺፺፺፺፺፺፺፺፻, ፺፺፺፺፺፺፺፺፺፺፺፺፺፩, ፺፺፺፺፺፺፺፺፺፺፺፺፺፪, ፺፺፺፺፺፺፺፺፺፺፺፺፺፫, ፺፺፺፺፺፺፺፺፺፺፺፺፺፬, ፺፺፺፺፺፺፺፺፺፺፺፺፺፭, ፺፺፺፺፺፺፺፺፺፺፺፺፺፮, ፺፺፺፺፺፺፺፺፺፺፺፺፺፯, ፺፺፺፺፺፺፺፺፺፺፺፺፺፰, ፺፺፺፺፺፺፺፺፺፺፺፺፺፱, ፺፺፺፺፺፺፺፺

Tokenization

Tokenization is one of the core tasks in working with text data, involving the process of breaking a long block of text into smaller units known as tokens [18]. These tokens can be words, letters, or subwords. After the text has been appropriately tokenized, further processing is typically carried out. In this study, tokenization was applied to Amharic texts using spaces or punctuation marks as delimiters. For instance, the Amharic sentence "ከችኮሽ" is tokenized into ["ከ", "ች", "ኮ", "ሽ"] using spaces as delimiters.

Table 1 Sample data normalization and cleaning of three Amharic sentences

Unnormalized and uncleaned data	Normalized and cleaned data
ሠላም ነው! Hello!	ሰላም ነው Hello
ኤችኤይቪ ከአንድ ሠው ወደ ሌላ ዕንዴት ይተላለፋል? How is HIV transmitted from one person to another?	ኤችኤይቪ ከአንድ ሰው ወደ ሌላ አንዴት ይተላለፋል How is HIV transmitted from one person to another
የኤችኤይቪ ኢንፌክሽን የተለመዱ ምልክቶች ምንድን ናቸው? What are the common symptoms of HIV infection?	የኤችኤይቪ ኢንፌክሽን የተለመዱ ምልክቶች ምንድን ናቸው What are the common symptoms of HIV infection

Table 2 Tokenization of "ኤችኤይቪ ኤድስ በሽታ ምንድን ነው? What is HIV AIDS disease?"

የኤችኤይቪ	ኤድስ	በሽታ	ምንድን	ነው
HIV(Human Imuno Virus)	AIDS(Aquired imuno deficiency Syndrum)	Disease	what	is

ኤድስ በሽታ ምንድን ነው?" can be tokenized into a list of words as shown in Table 2.

Feature extraction

After completing the text preprocessing activities, such as removing unnecessary symbols, punctuation, and normalizing and tokenizing it, the next step involves feature extraction. Machine learning algorithms cannot raw text directly, so a feature extraction method is applied to transfer the text into a feature matrix or vector [19]. This technique, known as word embedding or feature extraction, converts raw text data into a format suitable for input in to machine learning models. Feature extraction is a challenging aspect of text classification because irrelevant and overly frequent features can negatively affect the accuracy and performance of classifiers [20]. Research shows that various methods exist for extracting features from textual data, with Bag-of-Words (BoW), TF-IDF, FastText and Word2 Vec.

Bag of Words (BOW)

The Bag of Words (BOW) paradigm derives its name from treating each document as a "bag" containing its own unique set of words [21]. In this approach, each text document is represented as a numeric vector, where each dimension corresponds to a distinct word from the corpus, and the value indicates the word's frequency or occurrence in the document. This method enables the association of word frequency with specific documents. Commonly used for text categorization, BOW disregards the order of words and grammatical structure.

TF-IDF

TF-IDF (Term Frequency-Inverse Document Frequency) is a statistical method used to assess the importance

of a word within a document relative to a collection of documents [22]. It is calculated by multiplying a word's TF (Term Frequency), which measures how often a word appears in a document, by its IDF (Inverse Document Frequency), which reflects how common the word is across all documents. IDF is calculated as the logarithm of the total number of documents divided by the number of documents containing the word [23]. This method helps identify words that are frequent in a specific document but rare across the entire corpus, making them important for understanding the content of that document.

Word2 Vec

The Word2 Vec model represents a significant breakthrough in Natural Language Processing (NLP), enabling the extraction of meaningful features from text data. By converting words into numerical vectors, Word2 Vec captures semantic relationships and contextual information within a corpus. The model operates using two primary approaches: Continuous Bag-of-Words (CBOW) and Skip-Gram. CBOW predicts the target word based on its surrounding context. It uses the words around a given word to predict the target word, which makes it effective at capturing syntactic regularities and semantic relationships within a sentence [24]. However, CBOW can struggle with rare or out-of-vocabulary words since it relies solely on the local context for predictions. Skip-Gram, on the other hand, predicts context words based on a given target word. This approach excels at identifying more nuanced semantic relationships between words, making it particularly useful for capturing complex associations in larger datasets. However, due to its increased complexity, Skip-Gram often requires larger training datasets for optimal performance.

Fast text

FastText is a word embedding technique that has gained significant attention in the NLP, particularly for morphologically rich languages like Amharic. Unlike the traditional Word2 Vec model, FastText represents each word as a composition of character n-grams. A character n-gram is a continuous sequence of n characters or words, such as bigrams or trigrams. The word vector is computed as the sum of these character n-grams, enabling FastText to produce more effective embedding's, especially for rare words [25]. Since Amharic is a morphologically rich language with a complex word structure, and FastText can generate vectors for unseen words by using sub word information, in this study we used FastText techniques to convert textual data into numerical representations. This method effectively addresses variations in word forms and reduces the Out-Of-Vocabulary (OOV) problem, thereby enhancing the chatbot's ability to understand a divers inputs from users.

Modeling selection

Chatbot development may fall in two approaches i.e., either pattern matching or machine learning approaches [26]. Our focus is on machine learning chatbots, as the proposed chatbot is going to be developed on the principles of machine learning principles. Machine learning, a branch of Artificial Intelligence, enables a machine to learn from training data, recognize patterns, and make decisions with little human intervention or none human involvement. Since machine learning algorithms can be classified as supervised, semi-supervised, unsupervised, and reinforcement learning. Therefore, the selection of algorithms for a machine learning model depends on the type of training data and the intended activity. In our proposed model, we will use a dataset prepared in JSON format. In this dataset, the tag is going to be used as a class classifier. The patterns prepared for the user input is limited. So, our proposed model is going to be retrieval based chatbot. Retrieval based chatbots work on predefined in inputs and responses. The proposed model is going to have number of classes. Therefore we are about apply deep learning algorithms than machine learning algorithms as deep learning algorithms are efficient and easy to use with multiclass problems [27]. To compare machine learning and deep learning algorithms, findings indicate that deep learning provides higher accuracy in multiclass classification tasks. Therefore, our proposed work is used deep learning

techniques. Several Deep Neural Network models have been applied to NLP tasks. To develop our Amharic text based chatbot model for HIV/AIDS awareness and care advising we employed DL algorithms this model have mostly used Recurrent Neural Networks particularly LSTM, BiLSTM, GRU and BiGRU commonly used for text classification and summarization.

Model evaluation metrics

Model performance evaluation

After developing the model, it must be evaluated to determine whether the desired objectives have been achieved or not. Evaluating model performance provides measurable reflection into the learning process [28]. In this study, the proposed model is evaluated using different methods, incorporating metrics such as accuracy, recall, F1-score, and precision. Accuracy is the total correctly predicted data out of all training or testing data. It is used to measure how the model performed prediction of the given data into their corresponding classes. Accuracy is calculated by using Eq. 1 [29].

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

where, TP: True Positive: predicted values correctly predicted as actual positive.

FP: False Positive: Negative values predicted as Positive.

FN: False Negative: Positive values predicted as Negative.

TN: True Negative: predicted values correctly predicted as an actual Negative.

Precision answers how many are essentially correct from which are categorized as important. It expresses how many times the model predicts are true. It calculated using Eq. 2.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

Recall shows the number of proportions that are truly. It measures the completeness of the model. It calculated using Eq. 3.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

where, TP is true positive, and FN is False Negative

F-1 score is calculated as the weighted average of Precision and Recall. Hence, this score requires both false

positives and false negatives into consideration. But F1 is usually more suitable than accuracy, especially if we have an unbalanced class distribution. F-1 score evaluation metrics are used as evaluating intents in classification problems using Eq. 4 [30].

$$F1 - score = 2 * \frac{Precision + Recall}{Precision + Recall} \quad (4)$$

Human evaluation

The chatbot model should be evaluated to assess both its performance and user acceptance [31]. This can be accomplished by developing a graphical user interface (GUI) for the model, allowing end-users to interact with it directly. Based on user feedback, the study evaluates the model's accuracy by measuring how effectively it responds to user queries within the given domain.

Experiment, result and discussion

Data collection and preparation

We collected the data from various reliable and accessible platforms. The dataset for our research was gathered from several online websites, including the World Health Organization (www.who.int), the World Health Line and we also referenced various books relevant to this field. Furthermore, we compiled frequently asked questions from these online sources and conducted interviews with HIV patients and healthcare professionals to create a labeled dataset for training the deep learning model. The dataset consists of text scripts, which have been labeled in a question and response format according to identified intents. For labeling the data, we used the categories of intent, tag, pattern, and response. The tag represents the intent's class, the pattern corresponds to the questions, and the response is the output for the respective pattern in the dataset. Overall, we have 218 tags (or classes), 9426 patterns (or questions), and 647 distinct responses as presented in Table 3. Each tag contains specific patterns and responses, and the model's task is to classify a given pattern into its corresponding tag to generate the appropriate response associated with that class. Additionally, each pattern can have multiple responses based on the user's intent. After preparing the data, we used it for both training and evaluating the model. To do this, we first needed to access the data. As described in our data preparation

process, the dataset was created in JSON format and was accessed using the JSON library.

Implementation for data preprocessing

After loading the dataset, the next step was to preprocess the text data. Text preprocessing is an essential step to prepare data for use by machine learning algorithms. It ensures the data is complete, consistent, and ready for processing. we implemented various text preprocessing tasks, including tokenization, cleaning, and normalization. These preprocessing activities were applied to both the dataset and user input. Each of these tasks is discussed in detail one by one.

Text cleaning and normalization

The first step in text preprocessing is data cleaning, which involves removing unnecessary items such as punctuation, extra white spaces, and symbols from the dataset to improve the effectiveness of model training. Additionally, we applied text normalization, a technique that substitutes characters with others that have similar sounds, ensuring consistency in the data.

Tokenization

The process of transforming plain text into token sequence is called tokenization. The tokenization task results in a list of words by using spaces as delimiters. Different texts with varying representations are manually provided, and the experimental results are analyzed. The following code snippet illustrates how tokenization was implemented.

Hyperparameters

This section describes the parameters of our experimental study to train and test in the selected model. To achieve the intended results, we applied various experiments on different hyper parameters using both our training and testing data. Hyperparameters are settings that are chosen before training a model and are not learned from the data. The hyperparameters for the various model types (LSTM, BiLSTM, GRU, BiGRU) are consistent across the board, with an embedding dimension of 200, hidden dimension of 128, a batch size of 32, and a learning rate of 0.0001. All models utilize the Adam optimizer, SoftMax activation function, and Sparse categorical cross-entropy as the loss function, trained for 50 epochs with a dropout rate of 0.3.

Experimental result

Experimental results were conducted to evaluate the efficiency of the selected model by training it using various hyperparameters for the task. We used Google Colab to train the model, a platform that allows users

Table 3 The size dataset prepared for chatboat model

Dataset contents	Number
Tag	218
Patterns	9426
Responses	647

to write and execute Python code through a browser, primarily for data analysis and machine learning. Several deep learning models, particularly effective for NLP tasks, were tested during the experiments. The dataset consists of 218 tags, 9426 patterns, and 647 responses. We evaluated both simple recurrent model, such as LSTM and GRU, as well as more complex models with bidirectional layers, like BiLSTM and BiGRU. Bidirectional layers was explored to improve the model's ability to focus on important sequence elements during training. Different hyperparameters, such as learning rate, batch size, dropout and regularization techniques, were optimized to enhance performance and prevent overfitting. We applied different train-test split ratios: 70:30, 80:20, and 90:10. In the 70:30 split, 70% of the data was used for training, and 30% for testing. Similarly, the 80:20 split allocated 80% of the data for training and 20% for testing, while the 90:10 split used 90% of the data for training and the remaining 10% for testing. According to various studies, using a higher percentage of data for training generally leads to better model performance [30]. Adam optimizer with a learning rate (LR) of 0.0001 was used in this experiment as it provided the best performance compared to other optimizers and LR values. Evaluation metrics included Accuracy, Precision,

F1-score, and Recall. To reduce overfitting and enhance model performance, we used regularization techniques and early stopping in this study. Different RNN architectures were evaluated to determine their effectiveness. The tested models included LSTM, BiLSTM, GRU, and BiGRU.

Experimental results of LSTM model

In this case, we used hyperparameters such as batch size, embedding dimension, epochs, loss function, learning rate, and dropout to develop the LSTM model. Training was conducted for 50 epochs, with early stopping implemented to terminate the process once validation performance ceased to improve. The left side graph shows the training and validation accuracy, and the right graph shows the training and validation loss. The detailed results are presented in Fig. 2 and Table 4. As the model training, with an increase in the number of epochs, the accuracy rate for the training increased, and the loss decreased. We used Adam for the optimization to update network weights with a learning rate of 0.0001, achieving 93% accuracy for training and 87% for test. The experiment focused on measuring the model's performance using accuracy, precision, recall and F1 core.

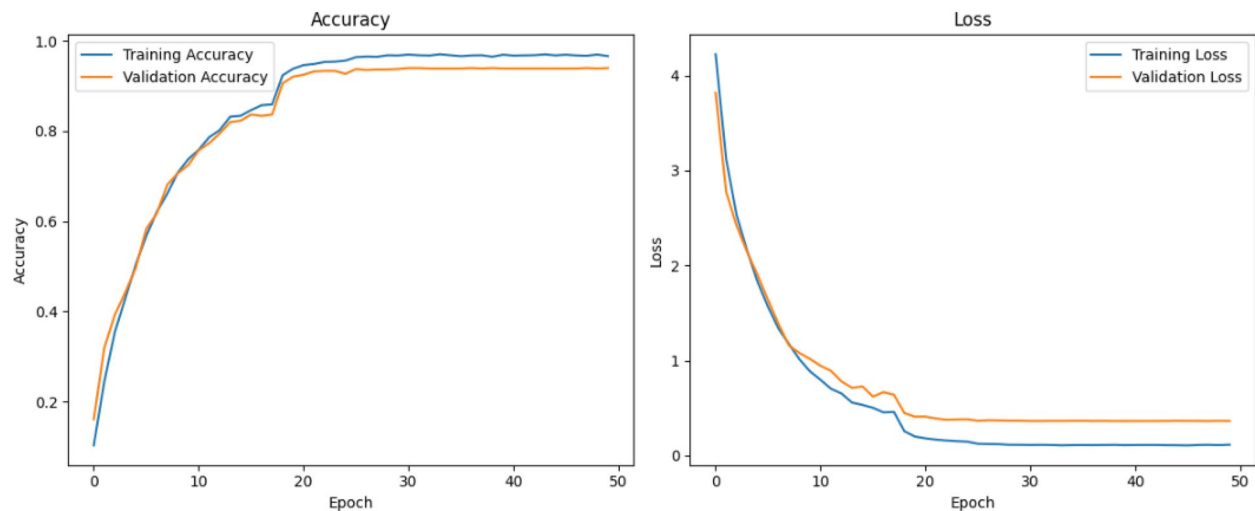


Fig. 2 LSTM model accuracy(left) and loss(right) graphs

Table 4 LSTM model results in different batch sizes

Algorithm	Bach size	Optimizer	Activation function	Performance metrics Results in percent			
				Accuracy	Precision	Recall	F1 score
LSTM	64	Adam	Softmax	86%	85%	84%	83%
	32	Adam	Softmax	87%	86%	85%	85%

Experimental results of BiLSTM Model

The next experiment was bidirectional LSTM. It achieved good performance in our experiments, demonstrating its capability to effectively capture contextual information in sequences. We used different hyperparameters such as, bach size, embedding dimension, epochs, loss function, learning rate, dropout and we used Adam optimizer. We

achieved 97% accuracy for training and 92% for test, and, the results of training and other performance evaluations are listed in Fig. 3 and Table 5 respectively.

Experimental results of GRU

As a result of applying the aforementioned hyper parameters, the experiment on GRU produced the following

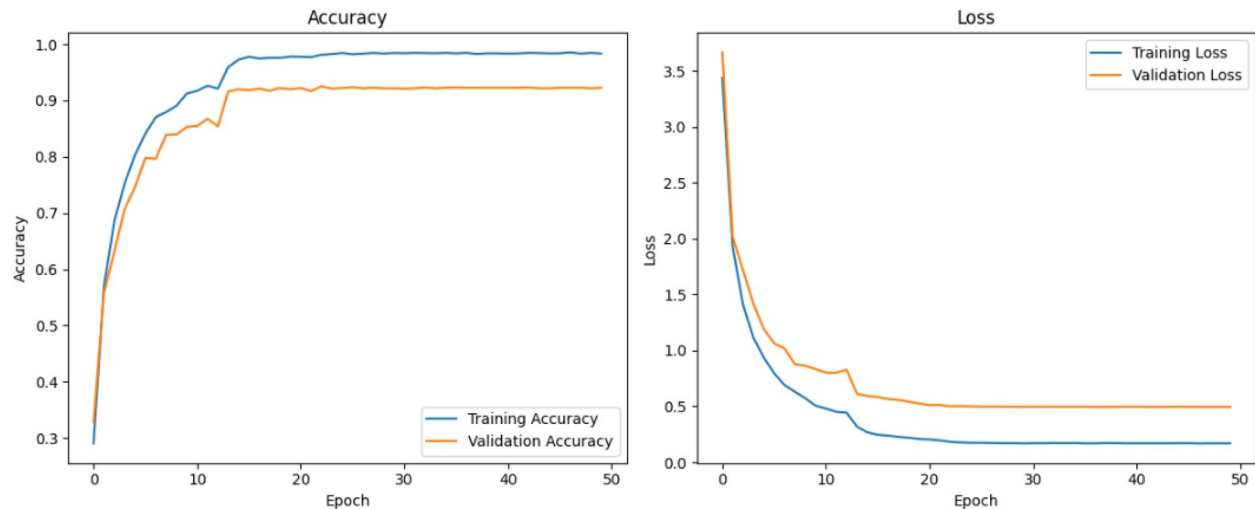


Fig. 3 BiLSTM model accuracy(left) and loss(right) graphs

Table 5 BiLSTM Model results in different batch sizes

Algorithm	Bach size	Optimizer	Activation function	Performance metrics Results in percent			
				Accuracy	Precision	Recall	F1 score
BiLSTM	64	Adam	Softmax	91%	90%	90%	89%
	32	Adam	Softmax	92%	91%	91%	90%

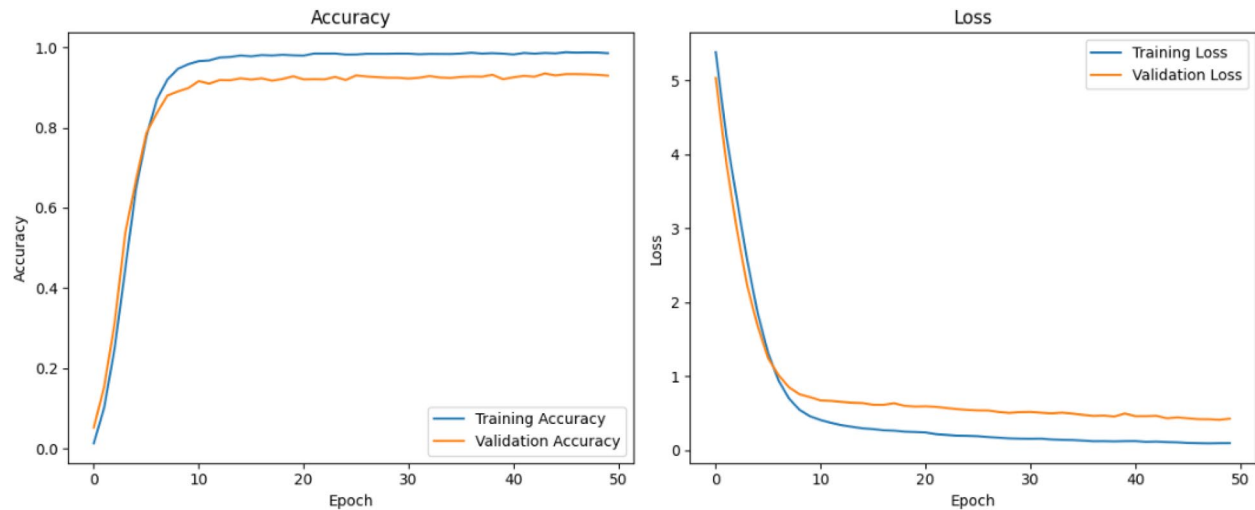


Fig. 4 GRU model accuracy(left) and loss (right) graphs

findings. The left side graph shows the training and validation accuracy, and the right graph shows the training and validation loss. Based on the evaluation metrics, the model's performance was assessed. The experiment that outperformed training accuracy and minimal loss with an average evaluation metrics was displayed. In Fig. 4. Suggests that the GRU model provides efficient learning with minimal overfitting during the training process.

In this experiment, the GRU model demonstrated efficient performance with faster convergence compared to the LSTM model. The final result showed a training accuracy of 98% and a test accuracy of 94%. And the result for other metrics are explored in the summarized Table 6.

Experimental results of BiGRU

We used BiGRU model to leverage context from both directions of the input sequence, enhancing the model's ability to understand and process conversational data.

Key parameters such as batch size, embedding dimension, epochs, dropout, and learning rate were set to optimize performance. The model was trained for a maximum of 50 epochs, with early stopping implemented to halt training when validation performance stopped to improve, preventing overfitting. As training progressed, the model's training accuracy showed stable improvement as indicated in Fig. 5, whereas, the training loss consistently decreased, this shows that the model was learning effectively. The accuracy value is not changed but, the curve of the train and test. The model's performance is further evidenced by its high achieving a test value of 95% as indicated in Table 7.

The BiGRU layers, with their ability to process sequences bidirectional, allowed the model to capture rich contextual information from both past and future inputs. The accuracy 98% of training, and of 95.01% of testing result.

Table 6 GRU model results in different batch sizes

Algorithm	Batch size	Optimizer	Activation function	Performance metrics Results in percent			
				Accuracy	Precision	Recall	F1 score
GRU	64	Adam	Softmax	93%	93%	92%	91%
	32	Adam	Softmax	94%	94%	93%	93%

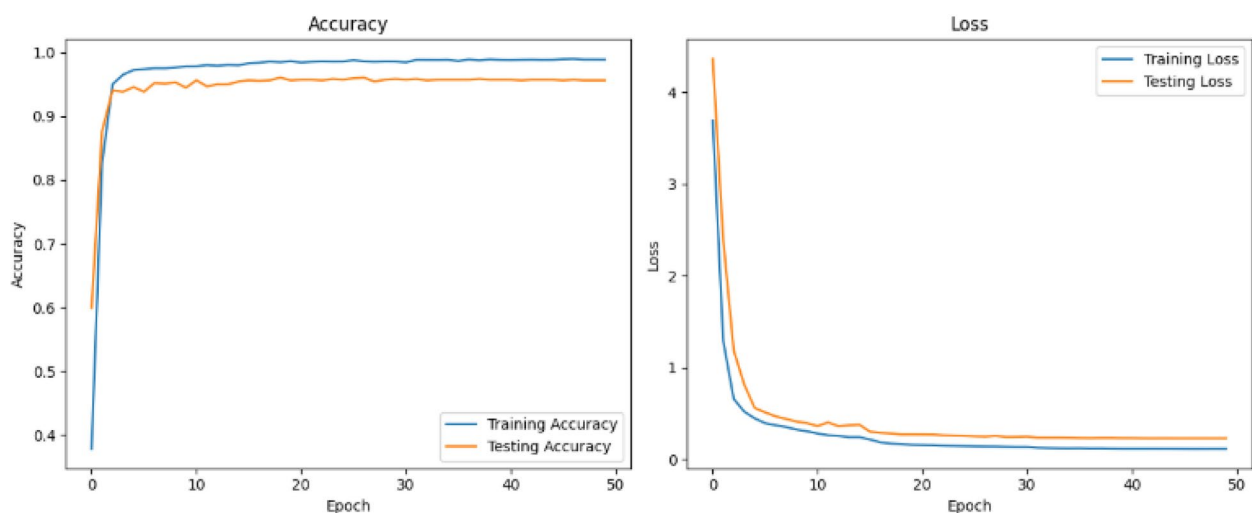


Fig. 5 BiGRU model accuracy (left) and loss (right) graph

Table 7 BiGRU model results in different batch sizes

Algorithm	Batch Size	Optimizer	Activation Function	Performance metrics Results in percent			
				Accuracy	Precision	Recall	F1 score
BiGRU	64	Adam	Softmax	94%	94%	93%	93%
	32	Adam	Softmax	95%	94%	93%	93%

Experiment results of LSTM, BiLSTM, GRU and BiGRU using different dataset split ratio

The above all experiments were done using 80:20 dataset split ratio. In this experiment we presented the training and testing accuracy history of our model for assisting HIV/AIDS patients, using a 90:10 and 70:30 split ratio. In this experiments, similarly, we used hyperparameters such as batch size, embedding dimension, epochs, loss function, learning rate, and dropout to develop models. With 50 epochs, and early stopping implemented to terminate the process once validation performance ceased to improve. The summarized experimental results are presented in Table 8.

Table 8 presented a detailed performance analysis of different recurrent neural network (RNN) architectures LSTM, BiLSTM, GRU, and BiGRU evaluated using two dataset split ratios: 70:30 and 90:10. The models are demonstrates based on performance metrics: accuracy, precision, recall, and F1 score. In summary, the results highlight that the BiGRU model consistently demonstrates best performance across both dataset splits, followed closely by GRU, BiLSTM, and LSTM. The metrics indicate that bidirectional architectures (BiLSTM and BiGRU) tend to perform better than their unidirectional counterparts (LSTM and GRU), with the BiGRU model being the most effective, using 80% for training and 20% for testing yields the best performance compared to the others.

Optimizer and activation function

To enhance the performance of the model, we explored the effects of different activation functions (ReLU and Softmax) and optimizer algorithms (Adam and SGD) across various RNN architectures, such as LSTM, BiLSTM, GRU, and BiGRU. This comparison is essential to identify the most effective combination of activation function and optimizer for increasing model accuracy and decreasing loss. In the comparison, we explored that the Adam optimizer consistently outperformed SGD across deep learning algorithms, including LSTM, BiLSTM, GRU, and BiGRU. Adam provided higher accuracy and lower loss due to its adaptive learning rate and momentum, which helped the model converge

faster and handle the complexity of the chatbot's multi-class classification task. For example, to examined the comparison, in BiGRU model, Adam achieved 95.01% of accuracy and a 0.372 loss, whereas, SGD's only achieved 92% accuracy, 91% of precision, 90% of recall, 90% Of F1 score, and 0.563 loss, This indicates that the Adam optimizer has a clear advantage in terms of performance. In activation functions, Softmax performed better than ReLU across all models, which outputs probability distributions, helped the model better differentiate between multiple classes. In GRU model, Softmax reached 95.01% accuracy with a 0.372 loss, while ReLU achieved 92.29% accuracy with a 0.442 loss. This makes Softmax the ideal activation function for this chatbot's classification tasks, additionally Adam proves to be the superior optimizer for enhancing performance.

Human evaluation

As part of our evaluation methods, the proposed model was also assessed by users. To perform this, we prepared a prototype of our model to users and evaluate the model based on its responses and calculated using Eq. 5. Equation 5 is used to measure the correctness of the chatbot model through human evaluation. Table 9 presents the results of the human evaluation of the model.

$$\text{Correctness} = \frac{\text{Total num of correctly answered by the model}}{\text{Total num of questions users ask}} * 100 \quad (5)$$

Table 9 Human evaluation results

No	Evaluator	No. of questions	Correctly response	Missed response	Correctness in percent
1	Person1	15	13	2	86.66%
2	Person2	20	17	3	85%
3	Person3	14	12	2	85.71%
4	Person4	15	14	1	93.3%
5	Person5	8	7	1	87.5%
Total average					87.6%

Table 8 LSTM, BiLSTM, GRU and BiGRU model results in different training and testing ratio

List of Recurrent neural network	Dataset split ratio							
	70:30				90:10			
	Accuracy	Precision	Recall	F1 score	Accuracy	Precision	Recall	F1 score
LSTM	84%	83%	83%	82%	85%	84%	83%	84%
BiLSTM	88%	87%	87%	86%	90%	89%	89%	89%
GRU	91%	91%	91%	90%	93%	92%	92%	92%
BiGRU	93%	92%	92%	91%	94%	93%	93%	92%

As explained in Table 9, the accuracy (correctness) of the model based on the human evaluation is shown. From the experiment, we concluded that the model achieved a testing result of 87.6% accuracy, which is considered a good result. The experiment also demonstrates that the model misses a few key queries, with 10 out of 72 questions failed to get the desired response.

Discussions of results

We used a testing dataset and various evaluation methodologies to measure the model overall performance. We have conducted different distinct experiments with different hyper parameters and training testing splitting ratios to distinguish the better performing model. Then, using evaluation measure, the average accuracy and loss was examined in the above experiments. In this study, all models were trained using our dataset with batch size 64 and 32, with Adam optimizer. The amount of training data determines the model's performance. When a large amount of training data is used, the model's performance improves, whereas using a limited dataset results in decreased performance. The experiment was conducted using different deep learning algorithms including, LSTM, BiLSTM, GRU, and BiGRU with varying hyperparameters.

In our experiments, we used the following hyperparameters and outperformed the best performance of our models: random state of 42, Adam optimizer, Softmax activation, regularization LR of 0.0001, batch size of 32, 50 epochs, 0.3 dropout, and 128 number of neurons. The Adam optimizer used in experiment provided a better learning rate compared to SGD. The random state was set to 42 to help regularize the model and minimize overfitting, ensuring that the graph remained stable.

In Experiment 1, LSTM model was used, achieving an accuracy of 87%, precision of 86%, recall of 86%, and an F1-score of 85%. Experiment 2 incorporated a bidirectional LSTM model, which further improved performance. It achieved an accuracy of 92%, precision of 91%, recall of 91%, and an F1-score of 90%. Experiment 3 incorporated a GRU model, yielding an accuracy of 94%, precision of 93%, recall of 92%, and an F1-score of 92%. Experiment 4 introduced a BiGRU model, which demonstrated the highest performance among all the experiments. It achieved an impressive accuracy of 95.01%, precision of 93%, recall of 93%, and an F1-score of 92%. In general, the results show that the bidirectional models achieved better performance than the basic LSTM and GRU models.

The BiGRU model demonstrated the highest performance among LSTM and GRU models primarily due to its unique capability to process information in both

forward and backward directions, effectively capturing contextual dependencies within input sequences. Unlike traditional LSTM and GRU architectures, which analyze data in a single direction, the bidirectional approach of BiGRU enables a deeper understanding of the temporal context in language, allowing the model to maintain relevant information from previous and subsequent words more effectively. This bidirectional nature enhances the model's ability to generate coherent and contextually relevant responses, making it particularly suitable for tasks that require nuanced comprehension, such as natural language processing applications in healthcare. Consequently, the BiGRU's superior performance metrics which is evidenced by higher accuracy, precision, recall, and F1-scores highlight its effectiveness in tackling complex language tasks while providing reliable outputs.

Conclusions and recommendation

We prepared 10,291 text data entries from different reliable sources, representing common HIV/AIDS symptoms, causes, preventions, and treatments. Then, after collected data, we converted it to a semi structured format, which is the JSON data format. After we prepared our dataset, to sure our dataset was consistence, complete and normalization process, we have applied natural language processing techniques to the dataset. After preprocessing our data, the next task is text feature extraction, i.e., our deep learning model will understand what we input for training purposes. We have used FastText technique for feature extraction. Different deep learning algorithms were tested, including LSTM, BiLSTM, GRU, and BiGRU. We conducted extensive experiments using data split ratios of 90:10, 80:20, and 70:30 for training and testing, respectively, to determine the optimal performance of the proposed model. Both the Adaptive Moment Estimation (Adam) optimizer and the Stochastic Gradient Descent (SGD) optimizer were employed, with the Adam optimizer yielding the best results. To address the challenge of model overfitting, dropout and early stopping techniques were applied. The highest performance was achieved using the BiGRU model, achieving a testing accuracy of 95.01% and a loss of 0.372. This result was achieved with an 80:20 train-test split, a batch size of 32, epoch 50 and a learning rate of 0.0001. The results demonstrate that the proposed model can effectively assist individuals affected by HIV/AIDS by providing support in Amharic language. The chatbot's performance was assessed through human evaluation, where random queries were input into the model. The chatbot demonstrated promising results by accurately identifying intents

when they were present in the repository and providing appropriate exception messages when they were not.

For future research, it is recommended to expand the dataset for model training, enhance natural language processing techniques tailored for Amharic, and consider integrating speech capabilities into the chatbot. By addressing these areas, the chatbot's accessibility and usability could be significantly improved, allowing users to interact through voice commands and thereby broadening its impact.

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Declaration of Helsinki

We confirm that our research adheres to the principles of the Declaration of Helsinki.

Clinical trial number

Not applicable.

Authors' contributions

Eshete Derb Emiru(PhD) wrote the manuscript and give result analysis. Mis Abebe Syntayehu edit the manuscript and do the experiment. Finally we all review the last manuscript.

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Data availability

The data that support the findings of this study are available at https://github.com/eshetecola/Developing-Amharic-Text-based-Chatbot-Model-for-HIV-AIDS-Awareness-and-Care/blob/main/HIV_chatbot_Dataset.json and from the corresponding author.

Declarations

Ethics approval and consent to participate

Our study was approved by the Debre Markos University College of Medicine and Health Sciences Institutional Research Ethical Review Committee, reference number RCSTTD/298/01/15. Informed consent was obtained from all participants.
Not applicable.

Consent for publication

We hereby grant our consent for the publication of our work and confirm that all co-authors agree to this decision.

Competing interests

The authors declare no competing interests.

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